



Levers

A lever is a bar that swivels on a fixed point or fulcrum and makes it easier to move a load. When you move your end further (the effort), the load at the other end moves a short way powerfully.



One type of lever works like a see-saw with the fulcrum between the load and the effort.



Another type places the load between the fulcrum and the effort (as on a wheelbarrow).



A third type of lever, shown by tongs, places the effort between the fulcrum and the load.

Machines

Machines make tasks easier. They reduce the effort you need to move something, or the time it takes. They work either by spreading the load, or by concentrating your efforts.



Wheel and axle

An axle goes through the centre of a wheel. Together they work as a simple rotating machine that makes it easier to move something from one place to another.

Axle



Gears

Gears are wheels with teeth that interlock so that one wheel turns another. They increase speed or force. Gears on a bicycle affect how much you must turn the pedal to spin the wheel.



The pedal turns a wheel, which turns a smaller wheel at a greater speed.



Hands on

Try walking straight up a hill and then zig-zag your way up. The winding path works like a simple machine. It increases the distance you walk, but decreases the effort you use.